# Introduction to Computer Control Systems, 5 credits, 1RT485

**Date:** 2025-08-26

**Teacher on duty:** Daniel Arnström

**Number of problems:** 5

**Allowed aid**: A non-programmable calculator and mathematical handbooks

| | | |
|---|---|---|
| **Preliminary grades:** | grade 3 | 15 points |
| | grade 4 | 21 points |
| | grade 5 | 26 points |

Some general instructions and information:

- Your solutions should be given in English.
- Write only on one side of the paper.
- Write your exam code and page number on all pages.
- Do not use a red pen.
- Use separate sheets of paper for the different problems (i.e. the numbered problems, 1–5).

*With the exception of Problem 1,* ***all your answers must be clearly motivated!*** *A correct but unmotivated answer will score reduced points!*

Best of luck!

# Formula sheet

## Laplace transform table

Table 1: Basic Laplace transforms

| $f(t)$ | $F(s)$ | $f(t)$ | $F(s)$ |
|---|---|---|---|
| unit impulse — imp$(t)$ | $1$ | $e^{-at}\sin(bt)$ | $\frac{b}{(s+a)^2+b^2}$ |
| unit step — step$(t)$ | $\frac{1}{s}$ | $e^{-at}\cos(bt)$ | $\frac{s+a}{(s+a)^2+b^2}$ |
| unit ramp — ramp$(t)$ | $\frac{1}{s^2}$ | $\sinh(bt)$ | $\frac{b}{s^2-b^2}$ |
| $t$ | $\frac{1}{s^2}$ | $\cosh(bt)$ | $\frac{s}{s^2-b^2}$ |
| $t^n$ | $\frac{n!}{s^{n+1}}$ | $\frac{1}{2b}t\sin(bt)$ | $\frac{s}{(s^2+b^2)^2}$ |
| $e^{-at}$ | $\frac{1}{s+a}$ | $t\cos(bt)$ | $\frac{s^2-b^2}{(s^2+b^2)^2}$ |
| $\frac{1}{a}(1-e^{-at})$ | $\frac{1}{s(s+a)}$ | $\frac{\cos(bt)-\cos(at)}{a^2-b^2}; (a^2 \neq b^2)$ | $\frac{s}{(s^2+a^2)(s^2+b^2)}$ |
| $\frac{1}{(n-1)!}t^{n-1}e^{-at}; (n=1,2,3...)$ | $\frac{1}{(s+a)^n}$ | $\frac{\sin(at)+at\cos(at)}{2a}$ | $\frac{s^2}{(s^2+a^2)^2}$ |
| $\sin(bt)$ | $\frac{b}{s^2+b^2}$ | | |
| $\cos(bt)$ | $\frac{s}{s^2+b^2}$ | | |

Table 2: Properties of Laplace Transforms

| | |
|---|---|
| $\mathcal{L}[af(t)] = aF(s)$ | $\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$ |
| $\mathcal{L}[f_1(t)+f_2(t)] = F_1(s)+F_2(s)$ | $\mathcal{L}\left[t^2 f(t)\right] = \frac{d^2}{ds^2}F(s)$ |
| $\mathcal{L}\left[\frac{d}{dt}f(t)\right] = sF(s)-f(0)$ | $\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}F(s), \quad n \in \mathbb{N}$ |
| $\mathcal{L}\left[\frac{d^2}{dt^2}f(t)\right] = s^2F(s)-sf(0)-f'(0)$ | $\mathcal{L}\left[f\left(\frac{t}{a}\right)\right] = aF(as)$ |
| $\mathcal{L}\left[\int f(t)\,dt\right] = \frac{F(s)}{s}+\frac{1}{s}\left[\int f(t)\,dt\right]_{t=0}$ | $\mathcal{L}\left[\int_0^t f_1(t-\tau)f(t)d\tau\right] = F_1(s)F_2(s)$ |
| $\mathcal{L}[f(t-a)] = e^{-as}F(s)$ | $\mathcal{L}[e^{-at}f(t)] = F(s+a)$ |

## Open-loop and sensitivity functions

$$G_o(s) = G(s)F_y(s), \qquad S(s) = \frac{1}{1+G_o(s)}, \qquad T(s) = 1-S(s) = \frac{G_o(s)}{1+G_o(s)}$$

## State-space forms and transfer function relations

- State-space form and transfer function

$$\begin{aligned} \dot{x} &= Ax+Bu \\ y &= Cx+Du \end{aligned} \quad \Rightarrow \quad \boxed{G(s) = C(sI-A)^{-1}B+D}$$

- Associated matrices

$$\mathcal{S} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \qquad \mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

- LTI system with transfer function

$$\boxed{G(s) = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{s^n + a_1 s^{n-1} + \cdots + a_n}}$$

  i) Observable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ -a_3 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ b_3 - a_3 b_0 \\ \vdots \\ b_n - a_n b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

  ii) Controllable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & -a_2 & -a_3 & \cdots & -a_n \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} b_1 - a_1 b_0 & b_2 - a_2 b_0 & \cdots & b_n - a_n b_0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

- Solution to state-space equation

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0$$

  can be written as

$$\boxed{x(t) = e^{At} x_0 + \int_0^t e^{A\tau} B u(t-\tau) d\tau}$$

- Observer system

$$\boxed{\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x})}$$

## Feedback control structures

General linear feedback in Laplace form:

$$\boxed{U(s) = F_r(s)R(s) - F_y(s)Y(s)}$$

Common control structures in this form.

- PID controller:

$$F_y(s) = F_r(s) = F(s) = K_p + \frac{K_i}{s} + K_d s,$$

  where $K_p, K_i, K_d \geq 0$

- State-feedback controller with observer:

$$\begin{aligned} F_r(s) &= \left(1 - L(sI - A + KC + BL)^{-1}B\right)\ell_0 \\ F_y(s) &= L(sI - A + KC + BL)^{-1}K \end{aligned}$$

## Bode's phase diagram

Consider the loop gain $G_o(s) = \mu \frac{\prod_{k=1}^{n_z}(1-\tau_k s)}{s^g \prod_{k=1}^{n_p}(1-T_k s)}$, where for simplicity $\mu > 0$. The phase of $G(i\omega)$ at a given frequency is

$$\arg\left(G_o(i\omega)\right) = -g\,90^\circ + \left[\sum_{k=1}^{n_z} \arctan\left(\tau_k \omega\right) - \sum_{k=1}^{n_p} \arctan\left(T_k \omega\right)\right] \frac{180^\circ}{\pi}$$

## Routh–Hurwitz stability criterion

Let $\Phi(s) = a_0 s^n + b_0 s^{n-1} + a_1 s^{n-2} + ...$ be the characteristic polynomial.

$$\begin{array}{ccc} a_0 & a_1 & ... \\ b_0 & b_1 & ... \\ \hline c_0 & c_1 & ... \\ \vdots & \ddots & \end{array}$$

A generic element of the third row, $c_k$, can be computed as

$$c_k = \frac{b_0 a_{k+1} - a_0 b_{k+1}}{b_0}$$

## Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period $T$ can be written in discrete-time as:

$$\boxed{\begin{aligned} x_k &= Fx_k + Gu_k \\ y_k &= Hx_k \end{aligned}}$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

## Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period $T$ can be written in discrete-time as:

$$
\boxed{\begin{aligned} x(k+1) &= Fx(k) + Gu(k) \\ y(k) &= Hx(k) \end{aligned}}
$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

**Bilinear transform formula** ($\alpha \in \{0, \frac{1}{2}, 1\}$)

$$
s = \frac{1}{T} \frac{1 - q^{-1}}{\alpha + (1 - \alpha)q^{-1}}
$$

# Problem 1: basic questions (6/30)

Answer only *'true'* or *'false'*. Each correct answer gives 1 point, each wrong answer gives $-1$ point. Minimum total points for Part A, B and C is 0.

## Part A

*Note:* Write 'skip' if your home assignment 1 score $\geq 6.5$

i) When designing a feedback controller, to reduce the overshoot one needs to reduce the phase margin.

ii) A phase margin of $30°$ is enough to guarantee the closed-loop stability.

iii) If the system is affected by an unmeasurable disturbance, a feed-forward disturbance compensator should be implemented to mitigate the effect of the disturbance.

**(3 p)**

## Part B

*Note:* Write 'skip' if your home assignment 2 score $\geq 6.5$

Consider the state-space system

$$\begin{cases} \dot{x} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \\ y = \begin{bmatrix} 1 & 0 \end{bmatrix} x \end{cases}$$

i) The system is observable.

ii) The poles of its transfer function representation are $-1$ and $-2$.

**(2 p)**

## Part C

*Note:* Write 'skip' if you attended/completed the fourth computer session (D4)

i) When discretizing a continuous-time controller, a sampling frequency twice as large as the crossover frequency eliminates the aliasing problem.

**(1 p)**

# Proposed solution to problem 1

## Part A

i) False

ii) True

iii) False

## Part B

i) False

ii) False

## Part C

i) False

## Problem 2 (6/30)

Consider the system described by the following transfer function

$$G(s) = \frac{1}{s-5}$$

Consider the proportional-derivative controller $F(s) = K + Ks$, where $K > 0$.

**a)** For what values of $K \in \mathbb{R}$ does the controller yield a closed-loop stable system?

**(3p)**

**b)** Pick $K = 20$ and consider a step reference signal $r(t) = 1 \cdot \text{step}(t)$. What is the steady state <u>input</u> ($\lim_{t\to\infty} u(t)$)?

**(3p)**

## Proposed solution to problem 2

**a)** The open loop transfer function is

$$G_o(s) = F(s)G(s) = K\frac{s+1}{s-5}$$

The closed-loop transfer function is

$$G_c(s) = \frac{G_o(s)}{1+G_o(s)} = \frac{K+Ks}{(K+1)s-5+K}$$

The closed-loop pole is $s = \frac{5-K}{K+1}$. Since the denominator is always positive ($K > 0$), the condition is $5 - K < 0 \implies K > 5$.

**b)** First, we note that for $K = 20$ the closed-loop system is stable, so we can apply the Final Value Theorem.

$$U(s) = \frac{F(s)}{1+F(s)G(s)}R(s) = \frac{20s+20}{1+\frac{20s+20}{s-5}} = \frac{(20s+20)(s-5)}{21s+15}R(s)$$

Applying the Final Value Theorem on $U(s)$ we can solve the exercise:

$$u(\infty) = \lim_{s\to 0} sU(s) = \lim_{s\to 0} \cancel{s}\frac{(20s+20)(s-5)}{21s+15}\frac{1}{\cancel{s}} = -\frac{100}{15}$$

## Problem 3 (6/30)

Need4Coffee, a (fictitious) company manufacturing coffee machines, is designing a control system to regulate the water flow rate $y(t)$ applying a voltage $u(t)$ to the water pump. A ideal pump is characterized by the nominal transfer function

$$G(s) = \frac{0.1}{1+s}.$$

The existing controller $F(s)$ yields a stable closed-loop system characterized by the (nominal) complementary sensitivity transfer function $T(s)$ shown in Figure 1.

**a)** Assume now that the pump is instead characterized by the transfer function $G^0(s) = 0.1\frac{s+10}{s+1}$. What is the multiplicative uncertainty[1] $\Delta(s)$?

**(3p)**

**b)** The same feedback controller $F(s)$ is applied to the real system $G^0(s)$. What can be said about the stability of this closed-loop system? Can we state that it is stable/unstable? Motivate well your answer, drawing on Figure 1 if needed.

**(3p)**

[1] Also known as multiplicative model mismatch.

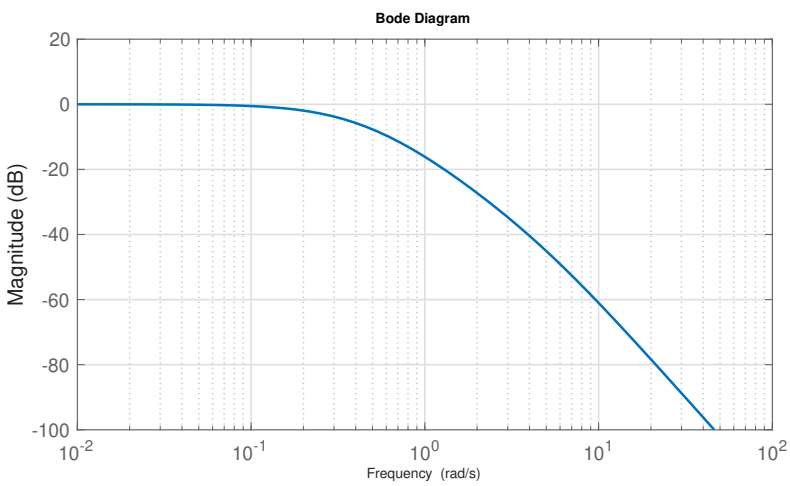


Figure 1: Magnitude Bode plot of the **nominal** complementary sensitivity transfer function $T(s)$.

## Proposed solution to problem 3

**a)** The model uncertainty $\Delta(s)$ is computed from the definition of multiplicative model uncertainty:

$$G^0(s) = G(s)(1 + \Delta(s))$$

Therefore

$$\Delta(s) = \frac{G^0(s)}{G(s)} - 1 = \frac{0.1\frac{s+10}{s+1}}{0.1\frac{1}{s+1}} - 1 = s + 9$$

**b)** To check whether $F(s)$ ensures the closed-loop stability of the real system $G^0(s)$, we can apply the criterion for the robustness to model mismatch (Result 6.2 of Glad & Ljung). To this end, we first compute $\frac{1}{\Delta(s)} = \frac{1}{s+9}$ and then sketch its magnitude Bode plot, $\left|\frac{1}{\Delta(i\omega)}\right|_{\text{dB}}$ onto Figure 1, resulting in Figure 2. Because at low frequencies the Bode plot of $\frac{1}{\Delta(s)}$ is below $T(s)$, closed-loop stability cannot be guaranteed.

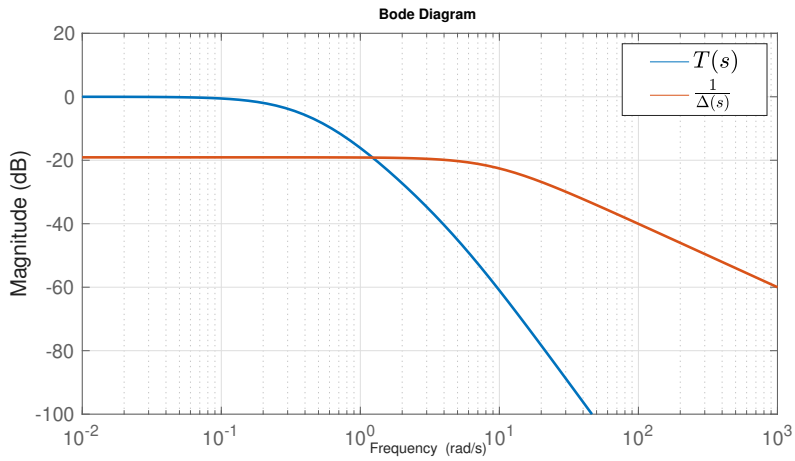


Figure 2: Sketch of the magnitude plot of $\frac{1}{\Delta(s)}$ (red line) compared to that of $T(s)$ from Figure 1.

## Problem 4 (6/30)

**a)** Consider a system described by the state-space model:

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

with matrices:

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

The system is to be controlled using state feedback, $u = -Lx$, where $L = \begin{bmatrix} l_1 & l_2 \end{bmatrix}$. Determine the values of $l_1$ and $l_2$ such that the closed-loop poles are placed at $s = -4$ and $s = -5$.

**(3p)**

**b)** For the same system, design a full-state observer. Determine the observer gain matrix $K = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$ such that the observer poles are placed at $s = -10$ and $s = -11$.

**(3p)**

## Proposed solution to problem 4

**a)** The characteristic polynomial of the closed-loop system is given by

$$\det(sI - (A - BL)) = \det \begin{bmatrix} s & -1 \\ 2 + l_1 & s + 3 + l_2 \end{bmatrix} = s^2 + (3 + l_2)s + (2 + l_1) = 0$$

The desired characteristic polynomial is

$$(s + 4)(s + 5) = s^2 + 9s + 20 = 0$$

Equating the coefficients gives

$$3 + l_2 = 9 \implies l_2 = 6,$$

$$2 + l_1 = 20 \implies l_1 = 18.$$

Thus, the controller gain matrix is $L = \begin{bmatrix} 18 & 6 \end{bmatrix}$.

**b)** The characteristic equation for the observer poles is

$$\det(sI - (A - KC)) = \det \begin{bmatrix} s + k_1 & -1 \\ 2 + k_2 & s + 3 \end{bmatrix} = s^2 + (k_1 + 3)s + (3k_1 + 2 + k_2) = 0$$

The desired characteristic polynomial is

$$(s + 10)(s + 11) = s^2 + 21s + 110 = 0$$

Equating the coefficients gives

$$k_1 + 3 = 21 \implies k_1 = 18$$

$$3k_1 + 2 + k_2 = 110$$

Substituting the value of $k_1$ into the second equation gives $k_2 = 52$.

Thus, the observer gain matrix is $K = \begin{bmatrix} 18 \\ 54 \end{bmatrix}$.

## Problem 5 (6/30)

**a)** A continuous-time controller is given by the transfer function $F(s)$ (from control error $e$ to the control $u$). In particular,

$$F(s) = 10\frac{s+2}{s+10}.$$

The controller is to be implemented digitally with a sampling period of $T = 0.1$ seconds. Using Tustin's method, determine the equivalent discrete-time controller transfer function, $F_d(q)$, where $q$ is the time-shift operator.

**(3p)**

**b)** Using the discrete transfer function $F_d(z)$ you found at point (a), derive the difference equation that relates the discrete control signal $u_k$ to the discrete error signal $e_k$. The difference equation should be of the form: $u_k = au_{k-1} + be_k + ce_{k-1}$ for some $a, b, c$.

**(2p)**

**c)** Assume the following sequence for the error signal $e_k$:

$$e_0 = 3.0, \quad e_1 = 1.5, \quad e_2 = 1.0.$$

If we assume the initial value of the control signal is $u_0 = 0$, what will be the control signal for the first two time steps? That is, what is the values for $u_1$ and $u_2$?

**(1p)**

## Proposed solution to problem 5

**a)** Tustin's method gives $s \approx \frac{2}{T}\frac{q-1}{q+1}$. Using this substitution in $F(s)$ gives

$$\begin{aligned} F_d(q) &= \frac{10\left(20\frac{q-1}{q+1}+2\right)}{20\frac{q-1}{q+1}+10} \\ &= \frac{22-18q^{-1}}{3-q^{-1}} \end{aligned} \tag{1}$$

**b)** We have that $u_k = F_d e_k$, which gives the difference equation

$$3u_k - u_{k-1} = 22e_k - 18e_{k-1}.$$

Solving for $u_k$ gives

$$u_k = \frac{1}{3}u_{k-1} + \frac{22}{3}e_k - 6e_{k-1}.$$

**c)** Direct insertion of the given values in the expression derived in b) gives

$$u_1 = -7, \qquad u_2 = -4.$$